

Novelty Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Novelty, Transience, and Resonance Analysis

PENPAL Project

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1 Overview

This notebook compares three information-theoretic metrics across conditions:

- **Novelty / surprise:** how unexpected a contribution is given prior context
- **Transience:** how quickly its influence fades
- **Resonance:** novelty minus transience

The notebook now follows the new data structure:

- it loads novelty scores through the shared comparison helpers,
- it restricts the analysis to **complete exchanges** in the aligned turn window,

- it treats `author_1` and `author_2` as **slot 1 / slot 2** rather than fixed identities,
- it uses role labels only where they are structurally valid:
 - **Human / AI** in Human-AI
 - **Starter-side / Non-starter-side** in same-type conditions
- role-labeled plots are therefore **within-condition descriptive views**, not direct cross-condition comparisons

Primary cross-condition object. Following the paper’s narrative-influence model, the main inferential result is the **within-condition novelty-to-resonance slope interaction** between the two slots, fit as $\text{resonance} \sim \text{surprise} \times \text{slot} + (1 \mid \text{conversation_id})$ per condition. The cross-condition asymmetry magnitude is

$$A_{\text{influence}}(\text{condition}) = \left| \beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}} \right|,$$

extracted via **emtrends**. Per-story OLS slope differences provide a label-invariant robustness layer. Bootstrap CIs, Cliff’s δ effect sizes, the planned HA vs (HH+AA)/2 contrast, and a length-controlled re-fit are added below. The signed Human/AI decomposition lives in `analysis/human-ai/novelty.Rmd` and is not duplicated here. Starter-mechanism analysis is deferred.

Metric-asymmetry companion. Because a story can also be asymmetric in the level of each information-theoretic metric, this notebook additionally computes

$$A_m(i) = \left| \bar{m}_{\text{slot=1}, i} - \bar{m}_{\text{slot=2}, i} \right|$$

for novelty/surprise, transience, and resonance. Each story therefore contributes one unsigned slot-gap magnitude per metric.

Article-use note. The article-facing result is the magnitude of the novelty-to-resonance slope gap. Do not infer Human-AI specificity from within-condition significance alone; the direct cross-condition question requires comparing the slot-slope gap across conditions and checking the per-story absolute gap as a robustness layer. Use the metric-asymmetry section when the claim concerns side differences in the metric levels themselves rather than the novelty-to-resonance slope.

2 Load Data

Table 1: Loaded novelty data by condition

condition	n_stories	n_turns
ai-ai	80	640
human-ai	97	677
human-human	36	288

3 Preprocessing

Slot-level rows: 3036

Metric rows: 9108

4 Primary: Within-Condition Slot Slope-Interaction Asymmetry

Table 2: Primary: $|\text{slope_1} - \text{slope_2}|$ from resonance $\sim$ surprise *
slot per condition

condition	slope_slot1	slope_slot2	signed_diff	se	t	p	A_influence	sig
Human-AI	1.090	1.060	-0.031	0.037	-0.824	0.410	0.031	
Human-Human	1.081	1.036	-0.046	0.066	-0.692	0.489	0.046	
AI-AI	0.951	0.959	0.008	0.039	0.213	0.831	0.008	

Table 3: Robustness: per-story $|\text{slope_1} - \text{slope_2}|$ (OLS per story,
4 points per slot)

condition	n_stories	median_A	mean_A
Human-AI	97	0.406	0.483
Human-Human	36	0.264	0.403
AI-AI	80	0.336	0.389

```
##
## Kruskal-Wallis on per-story A_influence:
##
## Kruskal-Wallis rank sum test
##
## data: A_influence_story by condition
## Kruskal-Wallis chi-squared = 2.9706, df = 2, p-value = 0.2264
##
##
## Pairwise Wilcoxon on per-story A_influence (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: per_story_slope_diff$A_influence_story and per_story_slope_diff$condition
##
## Human-AI Human-Human
## Human-Human 0.48 -
## AI-AI 0.28 0.86
##
## P value adjustment method: BH
```

5 Metric-Level Slot Asymmetry

Each story contributes one unsigned slot-gap magnitude per metric: $A_m(i) = |\bar{m}_{\text{slot } 1, i} - \bar{m}_{\text{slot } 2, i}|$. This compares asymmetry in the levels of novelty/surprise, transience, and resonance across conditions, separately from the novelty-to-resonance slope asymmetry above.

Table 4: Per-story $|\text{Slot 1 mean} - \text{Slot 2 mean}|$ by condition and
novelty metric

metric	condition	n_stories	mean_A	median_A	sd_A
surprise	Human-AI	97	0.828	0.826	0.490
surprise	Human-Human	36	0.355	0.340	0.218

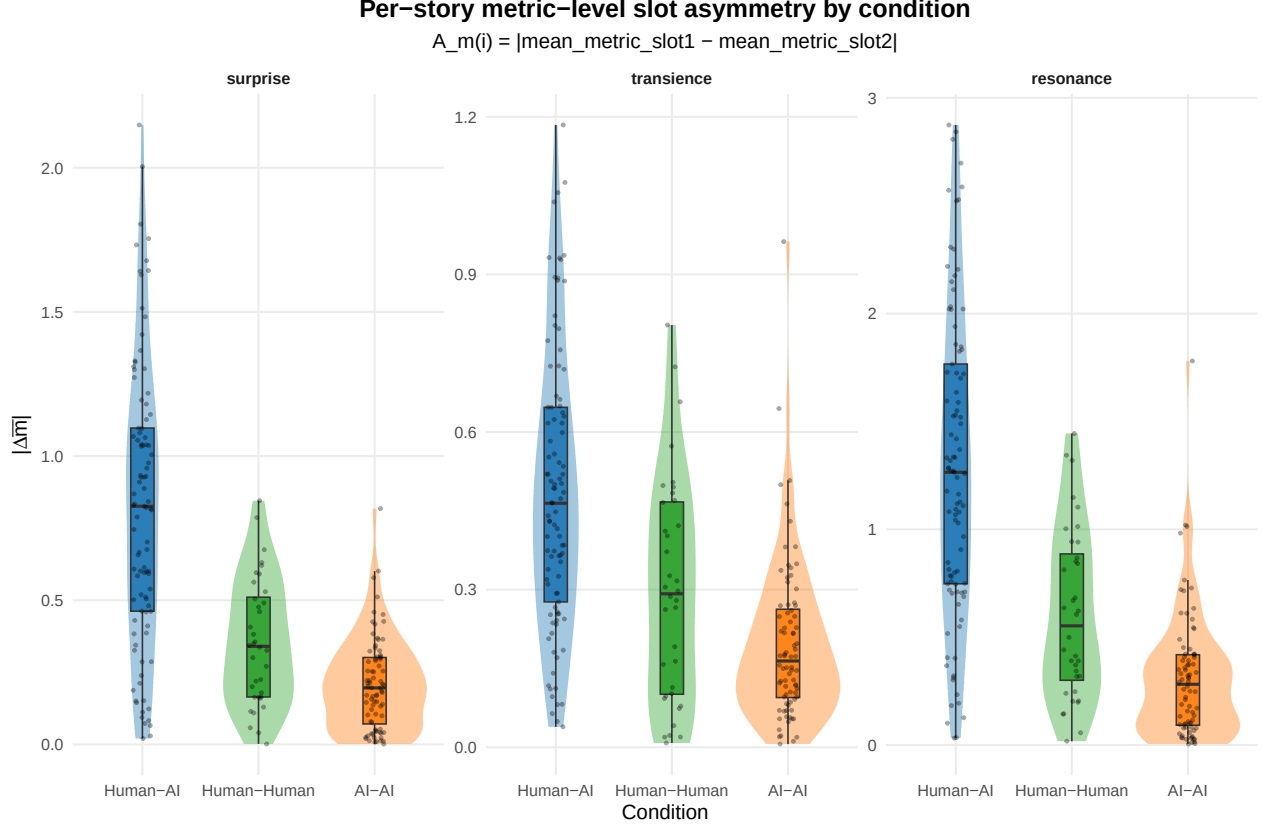
metric	condition	n_stories	mean_A	median_A	sd_A
surprise	AI-AI	80	0.209	0.196	0.160
transience	Human-AI	97	0.484	0.464	0.265
transience	Human-Human	36	0.300	0.292	0.213
transience	AI-AI	80	0.197	0.164	0.155
resonance	Human-AI	97	1.290	1.264	0.715
resonance	Human-Human	36	0.605	0.553	0.392
resonance	AI-AI	80	0.313	0.282	0.296

Table 5: Cross-condition tests for metric-level $A_m \sim \text{condition}$, BH-adjusted across metrics

metric	anova_p	kw_p	anova_p_BH	kw_p_BH
surprise	0	0	0	0
transience	0	0	0	0
resonance	0	0	0	0

Table 6: Pairwise Wilcoxon tests for metric-level A_m by condition, BH-adjusted within metric

metric	contrast	p_BH
surprise	Human-Human vs Human-AI	0.0000
surprise	AI-AI vs Human-AI	0.0000
surprise	AI-AI vs Human-Human	0.0006
transience	Human-Human vs Human-AI	0.0007
transience	AI-AI vs Human-AI	0.0000
transience	AI-AI vs Human-Human	0.0160
resonance	Human-Human vs Human-AI	0.0000
resonance	AI-AI vs Human-AI	0.0000
resonance	AI-AI vs Human-Human	0.0001



6 Bootstrap CIs and Effect Sizes on Metric-Level A_m

Table 7: Bootstrap 95% CI on per-condition mean of metric-level A_m ($R = 10,000$)

metric	condition	n	mean	ci_lower	ci_upper
surprise	Human-AI	97	0.828	0.732	0.926
surprise	Human-Human	36	0.355	0.284	0.428
surprise	AI-AI	80	0.209	0.175	0.245
transience	Human-AI	97	0.484	0.432	0.537
transience	Human-Human	36	0.300	0.233	0.370
transience	AI-AI	80	0.197	0.165	0.231
resonance	Human-AI	97	1.290	1.149	1.432
resonance	Human-Human	36	0.605	0.482	0.733
resonance	AI-AI	80	0.313	0.254	0.382

Table 8: Pairwise Cliff's delta on metric-level A_m

metric	contrast	n_a	n_b	cliff_delta	magnitude
surprise	Human-AI vs Human-Human	97	36	0.591	large
surprise	Human-AI vs AI-AI	97	80	0.766	large
surprise	Human-Human vs AI-AI	36	80	0.398	medium
transience	Human-AI vs Human-Human	97	36	0.397	medium
transience	Human-AI vs AI-AI	97	80	0.670	large

metric	contrast	n_a	n_b	cliff_delta	magnitude
transience	Human-Human vs AI-AI	36	80	0.281	small
resonance	Human-AI vs Human-Human	97	36	0.572	large
resonance	Human-AI vs AI-AI	97	80	0.806	large
resonance	Human-Human vs AI-AI	36	80	0.459	medium

7 Planned Contrast: Metric-Level HA vs (HH + AA)/2

Table 9: Planned contrast: metric-level A_m in HA vs the average of HH and AA, BH-adjusted across metrics

metric	contrast	estimate	se	t	df	p	ci_lower	ci_upper	p_BH
surprise	Human-AI vs (Human-Human + AI-AI)/2	0.546	0.054	10.158	124.201	0	0.44	0.652	0
transience	Human-AI vs (Human-Human + AI-AI)/2	0.236	0.033	7.058	148.304	0	0.17	0.301	0
resonance	Human-AI vs (Human-Human + AI-AI)/2	0.831	0.081	10.225	135.368	0	0.67	0.991	0

8 Bootstrap CIs and Effect Sizes on A_influence_story

Table 10: Bootstrap 95% CI on per-condition mean of per-story A_influence (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	97	0.483	0.416	0.555
Human-Human	36	0.403	0.317	0.492
AI-AI	80	0.389	0.327	0.457

Table 11: Pairwise Cliff's delta on per-story A_influence

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	97	36	0.112	negligible
Human-AI vs AI-AI	97	80	0.147	negligible
Human-Human vs AI-AI	36	80	0.021	negligible

9 Planned Contrast: HA vs (HH + AA)/2

Table 12: Planned contrast: per-story A_influence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.087	0.045	1.92	167.222	0.057	-0.002	0.176

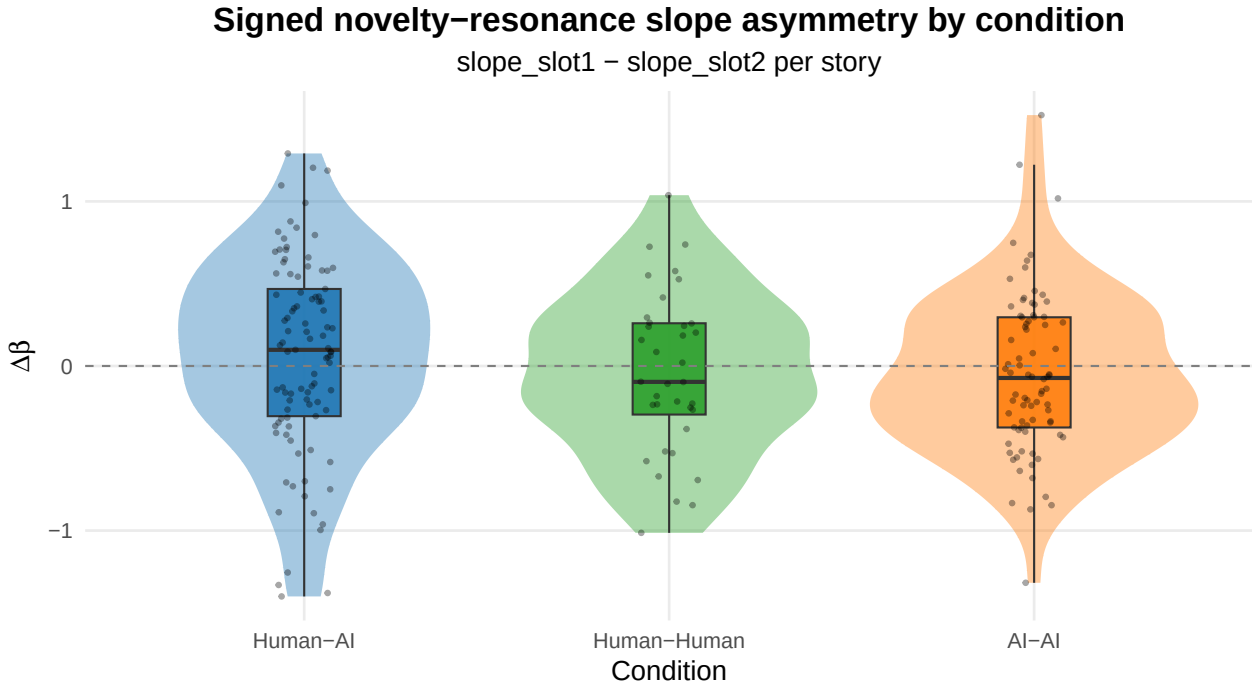
10 Directional Stability of Signed Novelty-Resonance Slope Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 novelty-to-resonance slope difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \beta_i^{\text{slot1}} - \beta_i^{\text{slot2}}$. HA-specific identity-tied directionality predicts high stability in HA and low stability in HH/AA, even when unsigned **A_influence** is similar.

Table 13: Per-condition directional stability of signed novelty-resonance asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	97	0.062	0.594	0.104	0.577	0.310
Human-Human	36	-0.040	0.490	0.082	0.528	0.624
AI-AI	80	-0.050	0.488	0.103	0.588	0.361

Brown-Forsythe variance-equality p = 0.1522



11 Length-Controlled Variant of A_influence

Re-fit the within-condition LMM with turn length as an additive covariate and recompute the unsigned slope-slot magnitude. Tests whether condition-level slot-asymmetry differences survive turn-length adjustment.

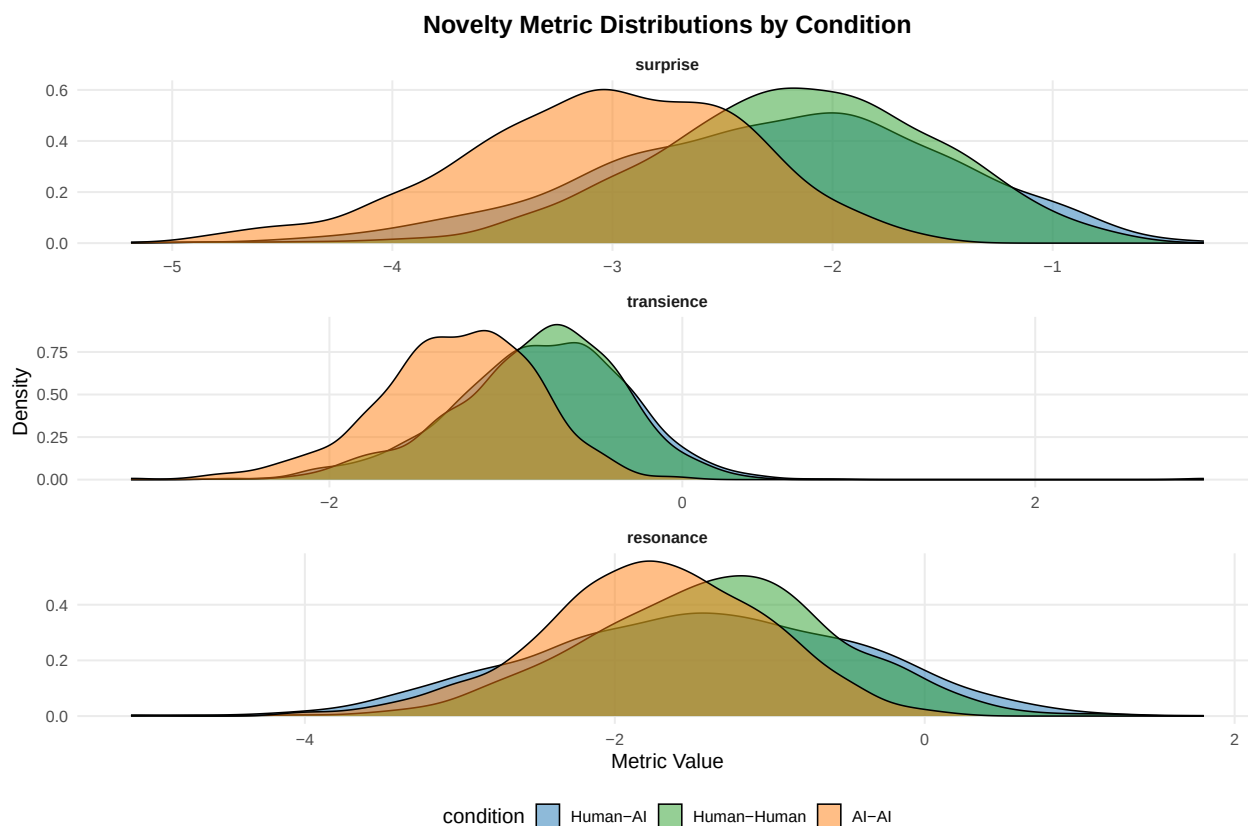
Table 14: Length-controlled $|\text{slope_slot1} - \text{slope_slot2}|$ (resonance ~ surprise * slot + nwords)

condition	A_influence_lc	se	p
Human-Human	0.043	0.066	0.511

12 Supplemental: General Condition-Level Comparison

Table 15: Overall novelty metrics by condition

condition	metric	n_stories	n_obs	mean_value	sd_value
Human-AI	surprise	97	1257	-2.272	0.778
Human-AI	transience	97	1257	-0.808	0.490
Human-AI	resonance	97	1257	-1.463	1.029
Human-Human	surprise	36	540	-2.169	0.639
Human-Human	transience	36	540	-0.811	0.492
Human-Human	resonance	36	540	-1.358	0.819
AI-AI	surprise	80	1239	-3.036	0.639
AI-AI	transience	80	1239	-1.282	0.453
AI-AI	resonance	80	1239	-1.754	0.717



```
##
## === surprise ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 6217.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```

## -4.0475 -0.6253 0.0387 0.6748 3.0554
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 0.07758 0.2785
## Residual 0.41280 0.6425
## Number of obs: 3036, groups: conversation_id, 213
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) -2.27251 0.03359 222.58156 -67.65 <2e-16 ***
## conditionHuman-Human 0.10309 0.06362 209.90621 1.62 0.107
## conditionAI-AI -0.76400 0.04931 211.17069 -15.49 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cndH-H
## cndtnHmn-Hm -0.528
## condtnAI-AI -0.681 0.360
##
## contrast estimate SE df z.ratio p.value
## (Human-AI) - (Human-Human) -0.103 0.0636 Inf -1.620 0.3155
## (Human-AI) - (AI-AI) 0.764 0.0493 Inf 15.494 <.0001
## (Human-Human) - (AI-AI) 0.867 0.0650 Inf 13.343 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === transience ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 4050.8
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -3.9352 -0.6112 0.0704 0.6432 8.1358
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 0.01661 0.1289
## Residual 0.20986 0.4581
## Number of obs: 3036, groups: conversation_id, 213
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) -0.808047 0.018391 233.105975 -43.937 <2e-16 ***
## conditionHuman-Human -0.003046 0.034471 210.884716 -0.088 0.93
## conditionAI-AI -0.473905 0.026746 213.035565 -17.719 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

##
## Correlation of Fixed Effects:
##      (Intr) cndH-H
## cndtnHmn-Hm -0.534
## condtnAI-AI -0.688  0.367

## contrast      estimate      SE  df z.ratio p.value
## (Human-AI) - (Human-Human)  0.00305 0.0345 Inf   0.088  1.0000
## (Human-AI) - (AI-AI)        0.47391 0.0267 Inf  17.719  <.0001
## (Human-Human) - (AI-AI)     0.47086 0.0350 Inf  13.441  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === resonance ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ condition + (1 | conversation_id)
## Data: filter(df_long, metric == metric_name)
##
## REML criterion at convergence: 7820.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.3214 -0.6305  0.0120  0.6642  3.7529
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.01314  0.1146
## Residual              0.75462  0.8687
## Number of obs: 3036, groups: conversation_id, 213
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -1.46368    0.02713 244.62921 -53.959  < 2e-16 ***
## conditionHuman-Human  0.10535    0.04998 206.79296   2.108   0.0362 *
## conditionAI-AI      -0.29052    0.03885 210.26522  -7.478 2.01e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndH-H
## cndtnHmn-Hm -0.543
## condtnAI-AI -0.698  0.379

## contrast      estimate      SE  df z.ratio p.value
## (Human-AI) - (Human-Human)  -0.105 0.0500 Inf  -2.108  0.1051
## (Human-AI) - (AI-AI)        0.291 0.0388 Inf   7.478  <.0001
## (Human-Human) - (AI-AI)     0.396 0.0504 Inf   7.861  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests

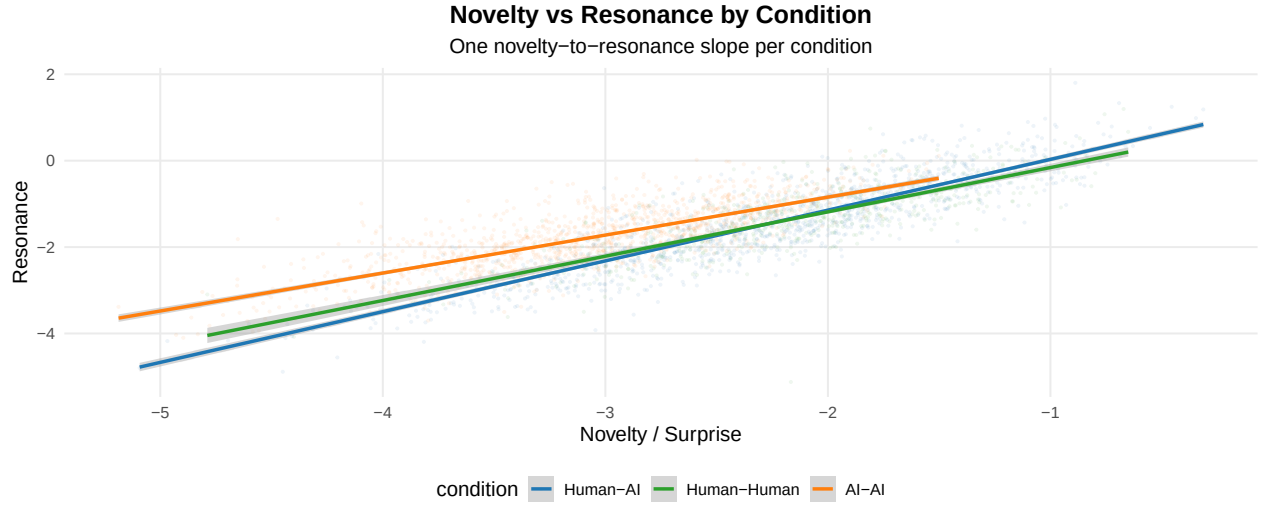
```

13 Supplemental: Pooled Novelty-to-Resonance Relationship

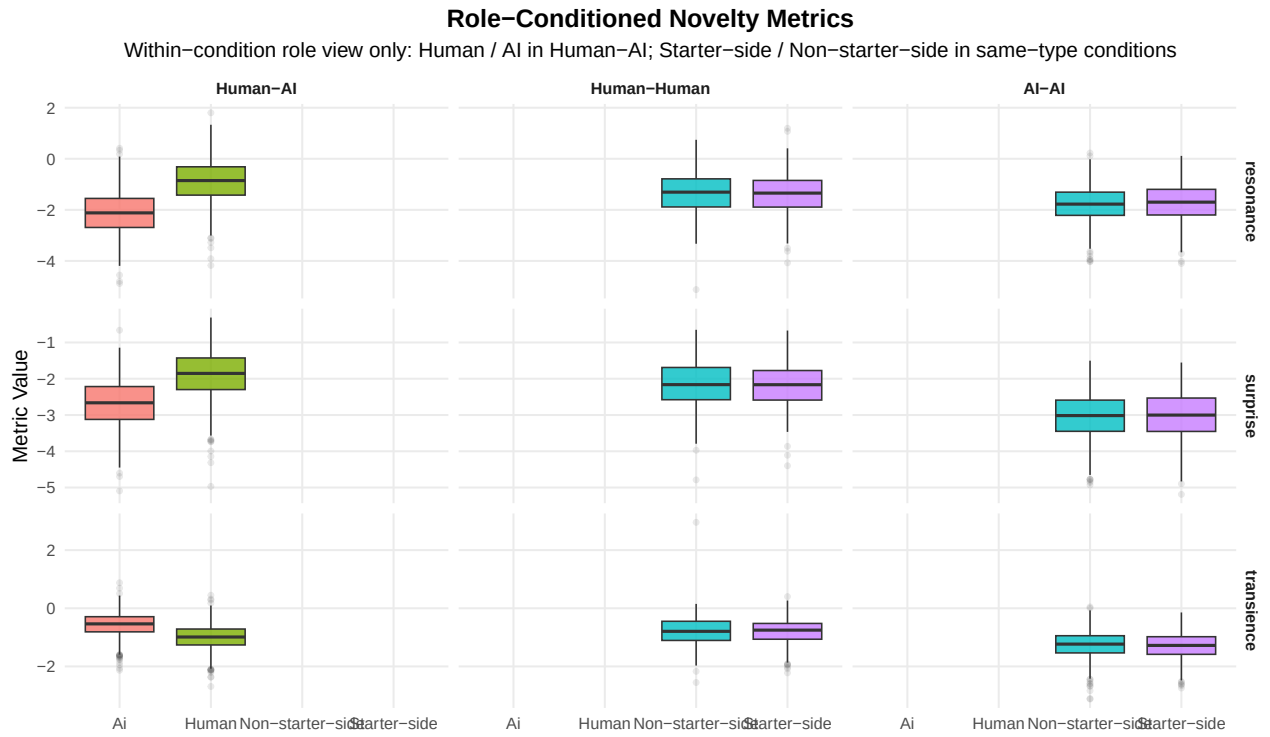
```
## Novelty-to-resonance model:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: resonance ~ surprise * condition + (1 | conversation_id)
## Data: df_slots
##
## REML criterion at convergence: 3902.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -8.4009 -0.6293 -0.0508  0.5941  3.8676
##
## Random effects:
## Groups             Name                Variance Std.Dev.
## conversation_id (Intercept) 0.02231  0.1494
## Residual                  0.19606  0.4428
## Number of obs: 3036, groups: conversation_id, 213
##
## Fixed effects:
##                                Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)                   1.30038    0.04294 1912.19610  30.283 < 2e-16
## surprise                      1.21667    0.01680 3027.99607  72.408 < 2e-16
## conditionHuman-Human          -0.34705    0.08761 1927.55370  -3.962 7.72e-05
## conditionAI-AI                -0.15158    0.08231 2047.58479  -1.842  0.0657
## surprise:conditionHuman-Human -0.15111    0.03623 3013.86694  -4.171 3.12e-05
## surprise:conditionAI-AI        -0.26051    0.02775 2907.88419  -9.389 < 2e-16
##
## (Intercept)                  ***
## surprise                     ***
## conditionHuman-Human         ***
## conditionAI-AI                .
## surprise:conditionHuman-Human ***
## surprise:conditionAI-AI      ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) surprs cndH-H cAI-AI sr:H-H
## surprise      0.889
## cndtnHmn-Hm   -0.490 -0.436
## condtnAI-AI   -0.522 -0.464  0.256
## srprs:cnH-H   -0.412 -0.464  0.906  0.215
## srprs:AI-AI   -0.538 -0.606  0.264  0.929  0.281
##
## Condition-specific novelty-to-resonance slopes:
## condition  surprise.trend      SE df asymp.LCL asymp.UCL
## Human-AI      1.217 0.0168 Inf    1.184    1.250
## Human-Human    1.066 0.0321 Inf    1.003    1.128
## AI-AI          0.956 0.0221 Inf    0.913    0.999
##
## Degrees-of-freedom method: asymptotic
```

Confidence level used: 0.95



14 Supplemental: Role-Level View



15 Summary

Primary cross-condition result: within each condition, $A_{\text{influence}} = |\beta_{\text{surprise, slot=1}} - \beta_{\text{surprise, slot=2}}|$ from $\text{resonance} \sim \text{surprise} * \text{slot} + (1|\text{conversation_id})$. The pooled LMM contributes only the unsigned magnitude column. Per-story OLS slope differences provide the label-invariant robustness test.

Metric-level companion result: each story also contributes $A_m(i) = |\bar{m}_{\text{slot } 1, i} - \bar{m}_{\text{slot } 2, i}|$ for surprise, transience, and resonance. These tests compare side-level asymmetry in the metrics themselves across conditions, rather than asymmetry in the novelty-to-resonance slope.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of per-story **A_influence**
- per-metric tests, bootstrap CIs, Cliff's δ , and HA vs (HH+AA)/2 contrasts for metric-level **A_m**
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2
- length-controlled re-fit (+ **nwords**) of the unsigned magnitude

Supplemental views retained:

- condition-level marginal distributions and pooled **condition** LMMs
- pooled novelty-to-resonance model with **surprise * condition** interaction
- within-condition role box-plots (descriptive only; HA Human/AI inferential view lives in analysis/human-ai/novelty.Rmd)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-11 23:28:48.314874